

Answer Key, Quiz 2
HS 239, 2026

6. Consider the mean variance model of investment in two risky assets under usual assumptions. Variance of asset 1 is 2 and that of asset 2 is 4. For diversification to work, we need the **covariance** is less than _____.

We have $\sigma_1 = \sqrt{2.0}$ and $\sigma_2 = 2$. Hence, we need $\rho < \min\left(\frac{\sigma_1}{\sigma_2}, \frac{\sigma_2}{\sigma_1}\right) =$
 $\min\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right) = \frac{1}{\sqrt{2.0}}$ (approx)
 However, by definition,

$$\begin{aligned}\rho &= \frac{cov(,)}{\sigma_1 * \sigma_2} < \frac{1}{\sqrt{2}} \\ \rightarrow cov(,) &< \frac{\frac{1}{\sqrt{2}} * 2}{\sqrt{2}} \\ \rightarrow cov(,) &< \frac{\sqrt{2}}{\sqrt{2}} = 1\end{aligned}$$